

MASMEC Budget Quote C610222-1

Modular All Drives Test System — Analysis & Summary

Supplier: MASMEC S.p.A. — Bari, Italy

Quote ref: C610222-1 | dated 2026-06-24

For: WEG Drives & Controls, Jaraguá do Sul (Santa Catarina), Brazil

Based on: WEG RFQ 2026-06-10, spec rev.00 (May 2026), meeting 2026-06-16

Scope: supply of **one (1) drive test line** to be installed at WEG's Santa Catarina plant. Base line = automated conveyor loop + 3 test modules (HIPOT, Functional, Load) + server. Total base budget **€1,250,000** (installation included). Manual load/wiring stations are a separate option.

1. Components / responsibilities WEG must supply

MASMEC's scope explicitly **excludes** the following, which it expects WEG to provide. These are the items to budget and plan for separately:

A. Equipment supplied by WEG into the line

Item	Where	Notes
Electronic loads (Load test)	OP40A — Load module	MASMEC supplies only the interface & coupling hardware to integrate the WEG electronic loads.
Electronic load (Functional test)	OP30A — Functional module	MASMEC also expects WEG to deliver an electronic load for the functional station, due to power-condition testing needs.
Master drives (OK / NOK)	Masters management trolley	Produced from series parts by WEG; quantity of OK/NOK masters still TBD.
Part samples	Design & set-up	Sample drives for tests/setup, shipped to MASMEC's Bari plant at WEG's transport cost.
Test fluid (if needed)	As required	Only if eventually necessary for tests.

B. WEG-brand components MASMEC will source & integrate

- **WEG PLC** — one per module (handles pallet IN/OUT and cycle start; the test routine itself runs on the PC).
- **WEG cMTX 15.6" HMI** operating panels — one per module/station.

(These are WEG products but bought/installed by MASMEC — listed for clarity, not a separate WEG delivery.)

C. Site, utilities & information WEG must provide

- **Power supply:** 380 Vac, 3-phase, 60 Hz to the control system; **compressed air** 5.5 bar.
- **Installation prerequisites** (excluded by MASMEC): unloading from transport, forklift/transpallet hire, positioning the machines, connection to electric/pneumatic/water mains, building work.
- **Remote assistance setup:** a company IP address with internet access plus the required HW/SW.
- **Technical data to reach a formal quote:** per-drive typology & versions, communication BUS types, connectors, 2D drawings of all drive models and interface features (HV/LV connectors, HMI, supports), detailed test specs per module, and HW/SW integration data for the WEG electronic loads.

2. How the pallets work

The base proposal includes **1 prototype pallet + 3 series pallets** plus all changeover tooling. The pallet is the product carrier that circulates the conveyor loop and docks into each test module.

- **Pallet base:** ≤ 400 mm × 480 mm, machined with 2 centring pins to receive changeover fixture nests.
- **Match plate** (STAUBLI Combitaq or similar) = the standardized pallet-side electrical interface: 3× AC IN + 3× AC OUT + signal connections (D-I/O, A-I/O, Bus, STO) + AC HV harness + LV connector for the changeover harness.

- **RFID TAG** on every pallet stores the product data; written at the load station after the drive QR code is read, then read at each test station (avoids extra QR readers downstream).
- **Changeover for 5 drive typologies:** 15 changeover fixture nests (5 types × 3 pallets) with model-specific centring pins and a QR code for typology check at loading; plus 15 changeover signal harnesses (5 types × 3 pallets).
- **Storage trolleys:** one for fixture types 1–3 (9 nests), one for types 4–5 (6 nests), and one for the master drives (2 parts).
- **Expandable:** extra pallets €6,000 each (Opt.3A); extra changeover set €4,000 per typology per pallet (Opt.3B).

In short: **each pallet is reconfigurable** across 5 drive models by swapping the fixture nest + harness, while the match plate keeps a single, repeatable electrical interface to the stations.

3. Wiring between product and pallet — manual vs automated

This is the most important distinction in the quote, and it is split into two separate connections:

Connection	Type	How it is done
Drive ↔ Pallet (loading & wiring the product onto the pallet)	MANUAL	Done by an operator at the optional manual stations OP10-M (load & wiring) and OP50-M (un-wiring & unload). Handling aided by a DEMAG chain manipulator. These stations are Option 1 (€150,000) , NOT in the base price. Harness assembly tools (screwdrivers etc.) are NOT included — extra cost.
Pallet ↔ Test module (match plate to station instruments)	AUTOMATED	Each test station has a pneumatic movement system + a connection unit (STAUBLI Combitaq) that automatically docks to the pallet match plate when the pallet is indexed in test position.

■ **Flag vs WEG's stated requirement.** WEG's concept called for an **automated electrical connection between the fixture and the product**. In this quote the **product-to-pallet wiring is manual** (operator stations); only the pallet-to-station connection is automated. If automated product-side wiring is required, it is not in this proposal and should be raised before the formal quote.

4. How the test modules work

Transversal "add-on" architecture: each module branches off the main conveyor flow via a 90° transverse conveyor unit + a 700 mm reversible segment, so a pallet moves IN/OUT of the station without blocking the main loop. New modules can be appended later (provided the conveyor was sized for it — see §5).

Common to every module

- Aluminium frame + polycarbonate panels (Load module uses thermal-insulation panels), steel base plate.
- Two pneumatic safety panels (open during pallet movement, closed during test), 3 door interlocks, emergency stops, Andon status light, air-treatment unit.
- Operator panel: WEG 15.6" cMTX HMI on a support arm. Cabinet: 380 Vac/60 Hz, WEG PLC for pallet IN/OUT + cycle start; **test routine fully executed by an industrial PC**.
- Pallet management: 3 pneumatic stops + 1 pneumatic index device; RFID read; pneumatic connection unit to the match plate.

OP20A — HIPOT module (€190,000)

- **Chroma 19020** multi-channel hipot tester (or equivalent), 10-channel for full AC / DC / IR testing; shielded HV leads; safety interlock & pass/fail.
- Hardware matrix for safe residual-charge discharge and automatic grounding at end of cycle.

OP30A — Functional module (€370,000)

- **DENSO 4-axis robot** (reach >200 mm) with a soft-terminal push tool to actuate the drive's HMI/pushbuttons without damage.

- Two vision cameras: drive HMI check + an additional vision check (subject TBD).
- **Siemens SINAMICS S120 grid simulator**: 220 / 440 / 575 V AC up to **60 A** from the 380 V/60 Hz supply (AC/DC + ALM DC/AC + active sinusoidal filter).
- Electronic load **supplied by WEG** (MASMEC adds interface/coupling).
- Voltage/current acquisition on 3 AC-IN + 3 AC-OUT channels via LEM probes (no power analyzer at this station).
- 3× communication BUS test gateways (Modbus RS485 / Ethernet / USB); chopper/BR resistor test; STO (Safe Torque Off) verification; ground-fault & output short-circuit protection test.

OP40A — Load module (€380,000)

- Thermally insulated cabin designed for up to **50 °C**, with a forced-convection heating unit + PID controller to hold the cabin at 50 °C.
- **Siemens SINAMICS S120 grid simulator**: 220 / 440 / 575 V AC up to **60 A** (same architecture as functional).
- Electronic load **supplied by WEG** (MASMEC adds interface/coupling).
- **Yokogawa WT5000** power analyzer + 7× 760902 input modules + 7× LEM current probes; acquisition of 3 AC-IN (L1,L2,L3) + 3 AC-OUT (U,V,W) + 1 DC.

■ **Configuration note.** This base quote includes **one of each** module (1 HIPOT, 1 Functional, **1 Load**). WEG's target throughput concept used **two Load modules**. A second OP40A would be an added module — MASMEC offers a **5% discount on module prices for each additional module purchased within 2027**.

5. Is the conveyor a fixed length?

Fixed in the base proposal, but intentionally extensible if specified up front.

- Base conveyor (OP00, €270,000): **BOSCH TS2** fly-roller chain loop — 9 m upper run (~900 mm height) + 9 m bottom return (~300 mm height), 2 electrical lifters closing the loop, 6 pallet stops.
- MASMEC explicitly notes **layout flexibility**: by defining the conveyor length larger than the base now, additional test modules can be added in a later stage thanks to the transversal add-on architecture.
- Individual stations (OP10/20/30/40/50) do **not** include a main-flow conveyor segment — the main conveyor is all in OP00.

Recommendation: if future expansion (e.g. a 2nd Load module or more typologies) is likely, oversize the conveyor in the first order. Retrofitting length later is far more disruptive than specifying it now.

6. Pricing & commercial summary

Pos.	Description	Budget (VAT excl.)
OP00	Conveyor system, 3 pallets, changeover equipment, masters management (9 m + 9 m)	€ 270.000
OP20A	HIPOT test module	€ 190.000
OP30A	Functional test module	€ 370.000
OP40A	Load test module	€ 380.000
S	Server (SQL DBMS, BOM, data collection, DataRelay)	€ 40.000
	TOTAL BASE — installation at Santa Catarina included	€ 1.250.000
Opt.1	Manual wiring & load/unload (OP10-M, OP50-M)	€ 150.000
Opt.2A	1 week on-site support (1 tech × 5 days, travel incl.)	€ 9.500
Opt.2B	1 week extension of on-site support	€ 6.500
Opt.3A	Additional pallet	€ 6.000
Opt.3B	Changeover equipment (1 typology, 1 pallet)	€ 4.000

- Base prices valid for simultaneous purchase at the first order; **5% discount per additional module bought within 2027**.
- **Delivery 50 weeks ARO**; transport **FCA Bari, Italy** (Incoterms 2020); installation max 15 working days; quote **valid 30 days**.
- **Warranty 12 months** from reception (max 14 from delivery); CE-certified (Machinery Directive 2006/42/EC); free remote assistance in warranty (MASMEC hours).
- **Import taxes excluded**; spare-parts list defined at end of design (optional budget available).
- Productivity commitment: line quality rate $\geq 99\%$, per-station technical availability $\geq 95\%$; line cycle time TBD by module quantity.

7. Key points to act on

- **Load module count**: base has only 1 Load module vs WEG's 2-module throughput concept — plan a 2nd OP40A (5% discount within 2027).
- **Manual product wiring**: product-to-pallet wiring is manual (Opt.1, €150k) — confirm whether automated product-side connection is actually required.
- **Conveyor sizing**: decide expansion intentions before order so the loop is sized for future modules.
- **WEG deliverables**: electronic loads (both Load & Functional), master drives, and sample parts are on WEG's side — assign owners and timing.
- **To get a binding (formal) quote**: provide per-drive specs, BUS types, connectors, 2D drawings, detailed per-module test specs, and electronic-load integration data.
- **Currency**: figures are in EUR, ex-works Bari, taxes excluded — convert to BRL and add freight/import duties for a true landed cost comparison.

Prepared as an internal reading of MASMEC budget quote C610222-1 for the Modular All Drives program. Figures and technical details are transcribed from the supplier's budget proposal and remain subject to MASMEC's formal quote.